CLOUD SERVICES

Sales & Profitability Analysis

Portfolio case study | Reporting period: 1 January – 31 December 2025

Business goal

Identify the largest revenue and profit pools, compare margin across services and markets, and turn transaction-level patterns into concrete, testable commercial actions.

Three key takeaways

1 | Scale and profitability are not identical

Data Analytics is the largest cloud service at 3.08M revenue (27.8% of total) and 1.74M profit, but its 56.62% margin is below the 57.07% portfolio margin.

2 | Enterprise is the largest segment, with mixed margin by service

Enterprise contributes 4.70M (42.5% of revenue). Within Enterprise, Data Analytics is the largest service-segment pool; Enterprise × SaaS Platform has the lowest margin of the 15 combinations.

3 | Focus the next decision on evidence, not volume alone

Cloud Storage × Mid-Market has the highest service-segment margin (58.59%) but only 4.64% of revenue. It is a candidate for a controlled growth test, not an automatic scale-up decision.

Key performance indicators

KPI	Validated result
Total revenue	11,072,461.41
Total profit	6,318,912.07
Aggregate profit margin	57.07%
Distinct Customer IDs	345

Dataset at a glance

Item	Details
Input dataset	1,512 raw rows × 8 columns; 1 Jan–31 Dec 2025.
Prepared analysis table	1,483 rows with 11 fields after adding Profit, Profit_margin and Month.
Cleaning	12 exact duplicate rows removed; 10 rows missing Revenue and 7 missing Cost after deduplication; 17 rows excluded in total.
Analytical unit	One cleaned row is treated as one recorded transaction for record-level metrics; customer/deal semantics are not independently verified.

Dashboard at a glance

The interactive Power BI report provides a compact view of overall performance, monthly revenue movement, service revenue and profit, segment and geographic mix, and margins by service and segment. Service-level margins span 56.1% to 57.9%; the small spread should be presented as context, not as proof of a material performance gap.

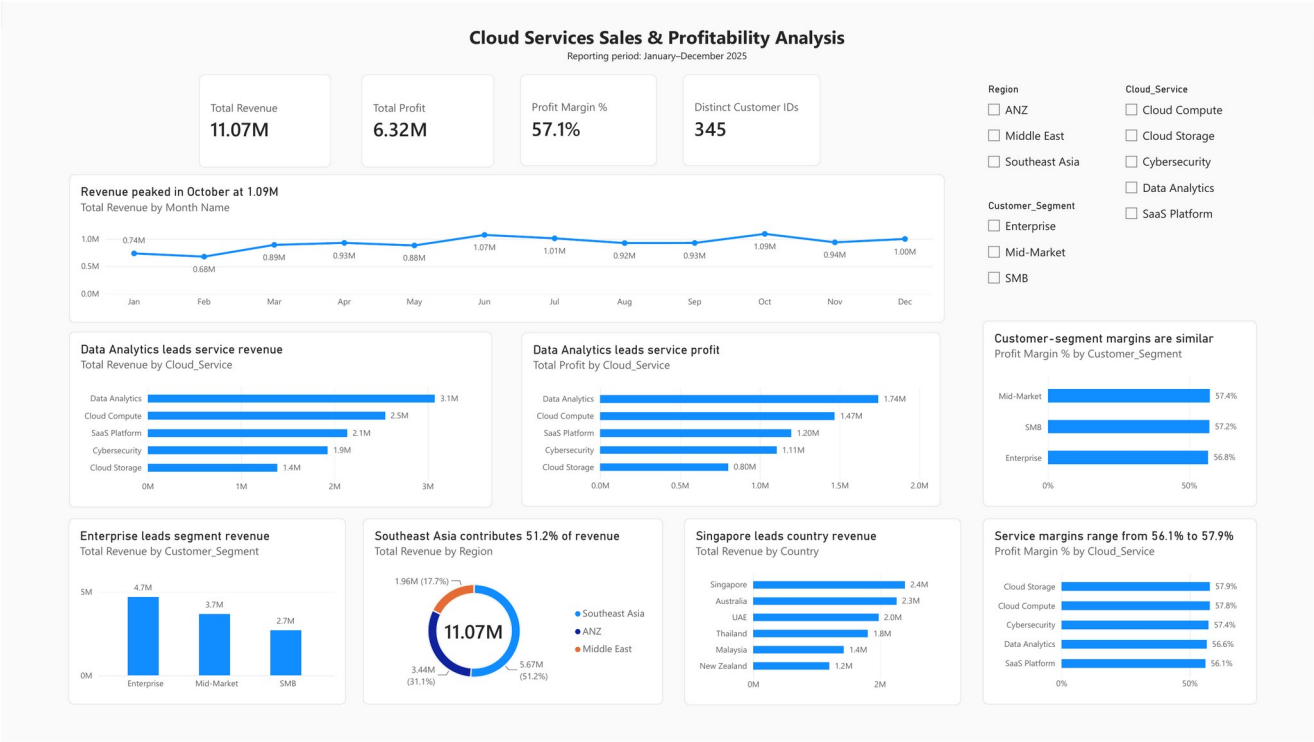


Figure 1. Updated Power BI dashboard snapshot (static export; the PBIX file is the interactive version).

What the report supports

- Compare the scale of five cloud services and three customer segments.
- Filter results by Region, Cloud_Service and Customer_Segment.
- Compare aggregate profit margin alongside revenue and profit; the country/region numbers should be treated as provisional until nine country–region mismatches are resolved.
- Read month-by-month revenue with markers, values and a zero baseline so visual differences are not exaggerated.

Reporting note

Amounts are shown in dataset units. The supplied file does not specify a currency symbol, so none is assumed.

Where revenue and margin sit

The portfolio margin is 57.07%. Service and segment margins vary, but their spreads are not equally large; focus on the combinations with both meaningful revenue and an actionable margin gap.

Performance by cloud service

Cloud service	Rows	Revenue	Rev. share	Profit	Margin
Data Analytics	311	3.08M	27.8%	1.74M	56.62%
Cloud Compute	344	2.54M	23.0%	1.47M	57.78%
SaaS Platform	308	2.14M	19.3%	1.20M	56.09%
Cybersecurity	226	1.93M	17.4%	1.11M	57.36%
Cloud Storage	294	1.39M	12.5%	0.80M	57.87%

Interpretation

Data Analytics contributes 27.8% of revenue, but Cloud Storage (57.87%) and Cloud Compute (57.78%) have the highest service-level margins. A growth decision should consider service scale and margin together.

Performance by customer segment

Segment	Rows	Revenue	Rev. share	Margin	Avg. revenue / row
Enterprise	375	4.70M	42.5%	56.76%	12,545
Mid-Market	530	3.67M	33.1%	57.36%	6,923
SMB	578	2.70M	24.4%	57.21%	4,669

Enterprise average recorded revenue per row is approximately 2.69× SMB (12,545 vs. 4,669). This describes recorded row values; it is not an average contract value or verified customer-level measure.

Profit margin by service × segment

Segment	Data Analytics	Cloud Compute	SaaS Platform	Cybersecurity	Cloud Storage
Enterprise	55.93% (n=91)	58.02% (n=81)	55.10% (n=85)	57.89% (n=55)	58.02% (n=63)
Mid-Market	57.70% (n=103)	57.40% (n=121)	56.40% (n=108)	56.87% (n=82)	58.59% (n=116)
SMB	56.61% (n=117)	57.93% (n=142)	57.61% (n=115)	57.12% (n=89)	56.60% (n=115)

Margin-screening rule used for the final recommendations: differences under 1.0 percentage point are treated as broadly similar; gaps of 1.0 point or more are flagged for diagnosis. This is a practical reporting heuristic selected during project review, not a preregistered threshold or a statistical significance test. Cell counts are shown as n. Enterprise × SaaS Platform is the lowest-margin cell (55.10%; n=85; 953,719 revenue); Mid-Market × Cloud Storage is the highest (58.59%; n=116; 514,259 revenue).

Market performance and data-quality checks

Regional results

Region as recorded	Rows	Revenue	Rev. share	Margin
Southeast Asia	815	5.67M	51.2%	57.00%
ANZ	443	3.44M	31.1%	56.94%
Middle East	225	1.96M	17.7%	57.49%

Country results

Country	Rows	Revenue	Rev. share	Margin
Singapore	259	2.39M	21.6%	56.96%
Australia	277	2.26M	20.4%	57.08%
UAE	227	1.98M	17.9%	57.39%
Thailand	306	1.81M	16.4%	57.95%
Malaysia	248	1.43M	12.9%	56.10%
New Zealand	166	1.20M	10.8%	56.55%

Commercial interpretation

Southeast Asia is the largest region as currently labelled (51.2% of revenue). Singapore leads country revenue (2.39M); Thailand has the highest country margin (57.95%), while Malaysia has the lowest (56.10%). Geographic recommendations remain provisional until the mapping checks below are resolved.

Customer-ID and region validation

Validation	Result
Distinct Customer IDs	345; show as IDs, not verified unique customers.
IDs appearing across multiple countries	315 of 345 IDs.
IDs appearing across multiple segment labels	300 of 345 IDs.
IDs appearing across multiple regions	287 of 345 IDs.
Country–region mapping check	9 of 1,483 rows (0.61%) have a region inconsistent with the country-to-region mapping used for QA; they represent 62,662.60 revenue (0.57% of total revenue).

Action: confirm customer-ID semantics and correct or explicitly document the nine country–region mismatches before using these results to allocate market budgets. Regional totals above reflect the supplied Region field as recorded.

Monthly movement and transaction-level signals

The data covers one calendar year. It shows movements within 2025, but does not establish recurring seasonality or explain the causes of individual increases and declines.

Monthly revenue and month-over-month change

Month	Revenue	Change vs. prior month	MoM growth	Rows
Jan	735,474.26	—	—	129
Feb	676,718.64	-58.8K	-7.99%	103
Mar	891,204.66	+214.5K	+31.70%	131
Apr	927,382.79	+36.2K	+4.06%	124
May	880,732.91	-46.6K	-5.03%	122
Jun	1,072,440.90	+191.7K	+21.77%	138
Jul	1,010,252.85	-62.2K	-5.80%	135
Aug	924,186.62	-86.1K	-8.52%	118
Sep	926,300.43	+2.1K	+0.23%	121
Oct	1,091,352.13	+165.1K	+17.82%	133
Nov	937,532.28	-153.8K	-14.09%	113
Dec	998,882.94	+61.4K	+6.54%	116

Monthly finding

October is the highest-revenue month (1,091,352.13); February is the lowest (676,718.64). Revenue rose 31.70% in March and 17.82% in October, then fell 14.09% in November. Check pipeline, timing, service mix and transaction counts before attributing causes.

Record-level distribution (not customer-level concentration)

Measure	Result
Average recorded revenue per row	7,466.26
Median recorded revenue per row	6,281.81
90th-percentile revenue threshold	13,369.41
Rows at or above the 90th-percentile threshold	149 rows; 22.72% of total revenue.

For this project, one cleaned row is treated as one recorded transaction. The top-decile statistic describes row-level revenue concentration; it must not be described as top-10-customer concentration because the customer IDs are not validated for that purpose.

Prioritized recommendations

Each recommendation links observed evidence to a specific action and a measure. Expected benefits are hypotheses to test; no realized business impact is claimed.

1. Grow the largest revenue pool with margin guardrails

Evidence: Data Analytics × Enterprise generated 1,445,685.95 revenue (13.06% of total) across 91 rows at 55.93% margin, 1.14 percentage points below the overall margin.
Action: prioritize qualified Enterprise Data Analytics opportunities, but require a deal-level cost and discount review before scaling; request missing discount and granular cost data rather than assuming it is available.
Measure: revenue, profit, margin and row count for this combination; scale only if a controlled test improves profit without further margin dilution.

2. Diagnose the weakest-margin combination

Evidence: Enterprise × SaaS Platform generated 953,718.90 revenue across 85 rows at 55.10% margin, the lowest of the 15 service × segment combinations.
Action: request discount-level and detailed cost data for this combination, then test one pricing or cost change at a time; the current dataset does not contain discount detail or granular cost components.
Measure: compare revenue, total profit, margin and row count against a defined baseline after each controlled change.

3. Test controlled growth in a higher-margin cell

Evidence: Mid-Market × Cloud Storage generated 514,259 revenue at 58.59% margin, the highest cell margin, representing 4.64% of total revenue.
Action: pilot a focused cross-sell or bundle for suitable Mid-Market prospects; confirm offer fit before broad rollout.
Measure: incremental revenue, profit and margin against a defined baseline or comparison.

4. Validate market mapping, then test focused expansion

Evidence: Southeast Asia represents 51.2% of revenue as currently labelled. Singapore and Australia together generated 4,656,527.63, or 42.06% of total revenue based on the unrounded combined totals. Nine country–region rows need validation.
Action: correct or document the nine mappings, then prioritize country × service opportunities using revenue share, margin and row count; run a small, time-bound test before shifting substantial resources.
Measure: track revenue, profit, margin, row count and data completeness monthly against a baseline.

Suggested 30 / 60 / 90-day sequence

Period	Work	Decision check
Days 1–30	Correct data mappings, confirm ID rules and establish revenue/margin baselines by service, segment, country and region.	Agree on baseline metrics and quality checks before acting.
Days 31–60	Run one small, time-bound commercial test (e.g., a targeted offer or cross-sell) with a defined audience and budget.	Track incremental profit and margin as well as revenue.
Days 61–90	Compare test performance with the baseline or a suitable comparison group. Document what worked and what did not.	Scale, adapt or stop using measured results; do not claim impact before the test.

Method, technical proof and limitations

Workflow and tools

- Python / pandas: inspected duplicate and missing-value conditions; parsed dates; standardized country and segment labels; calculated Profit = Revenue − Cost, profit margin and month; summarized service, segment, country and monthly performance.
- PostgreSQL: tested aggregate KPI queries, service/segment/region/country summaries, margins, and month-over-month changes. The main totals reconcile with the Power BI KPI cards.
- Power BI: KPI cards, monthly revenue trend, revenue/profit by service, revenue by segment/region/country, margin by service and segment, and slicers for Region, Cloud_Service and Customer_Segment.

Selected SQL examples

Margin by service and segment

```
SELECT "Customer_Segment", "Cloud_Service", COUNT(*) AS row_count,
    SUM("Revenue") AS revenue, SUM("Profit") AS profit,
    SUM("Profit") / NULLIF(SUM("Revenue"), 0) * 100 AS margin_pct
FROM public.cloud_saas_sales_clean
GROUP BY "Customer_Segment", "Cloud_Service"
ORDER BY revenue DESC;
```

Month-over-month growth

```
WITH monthly AS (
  SELECT "Month" AS month_no, SUM("Revenue")::numeric AS revenue
  FROM public.cloud_saas_sales_clean
  GROUP BY "Month"
), monthly_lag AS (
  SELECT month_no, revenue,
      LAG(revenue) OVER (ORDER BY month_no) AS prev_revenue
  FROM monthly
)
SELECT month_no, ROUND(revenue, 2) AS revenue,
    ROUND(
      (revenue - prev_revenue) / NULLIF(prev_revenue, 0) * 100,
      2
    ) AS mom_growth_pct
FROM monthly_lag
ORDER BY month_no;
```

The tested output showed October at +17.82% month over month and November at −14.09%.

Data limitations and release conditions

- Data-quality note: The simulated practice dataset contains duplicate rows, missing financial values, inconsistent category labels and Customer_ID mappings, plus nine country–region mismatches. This project demonstrates how these observed issues are cleaned or flagged; their individual causes are not inferred.
- Project scope: This independent portfolio project uses a simulated cloud-services sales dataset to demonstrate data cleaning, SQL analysis, Power BI dashboard development, and business insight generation. The findings illustrate the analytical approach and do not represent the performance of an actual company.
- Customer_ID does not function as a stable unique-customer identifier in this dataset: 315 of 345 IDs (91.3%) appear across multiple countries, 300 across multiple segment labels, and 287 across multiple regions. Customer-level concentration, repeat-purchase, retention, churn and LTV claims are therefore excluded.
- Nine rows have a country–region assignment inconsistent with the simple country-to-region mapping used for QA. Regional results are shown as recorded and should be refreshed after these mappings are resolved.
- The dataset spans one calendar year. It supports describing 2025 movement, but not proving recurring seasonality or the cause of revenue changes.
- The dataset has Revenue and Cost but does not include granular discounts, unit costs, pricing rationale, sales pipeline or campaign attribution. Therefore recommendations are hypotheses to test; no causal business impact is claimed.
- Currency is not specified in the provided file; figures are reported in dataset units. One row is used as one recorded transaction only for row-level analysis.

Conclusion

The analysis establishes where recorded revenue and profit are concentrated, highlights margin variation by service–segment combination, and identifies concrete priorities for further testing. The next business step is to validate the flagged data-quality items, then pilot the targeted actions with pre-agreed measures of revenue, profit and margin.

Portfolio and contact

Available services: sales data analysis, SQL reporting, Power BI dashboards and monthly performance reporting. Portfolio contact: GitHub profile - https://github.com/JessJiDataAnalyst.